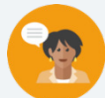


## 5.10 Congruence in Two-Dimensional Figures – Part 2

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                           |                                                                                  |                                                                                                                                                                                                                                      |                                                                                                       |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.                                                                                                                                                   |                                                                                  |  Learning Target                                                                                                                                  | <ul style="list-style-type: none"><li>I can solve real-world problems involving congruence.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                               |                                                                                  | Working Toward                                                                                                                                                                                                                       |                                                                                                       |
|                                                                                                       | N/A                                                                                                                                                                                                                                                                       |                                                                                  | N/A                                                                                                                                                                                                                                  |                                                                                                       |
|  Essential Question  | <ul style="list-style-type: none"><li>How can I use my knowledge of congruence of two figures to solve real-world problems?</li></ul>                                                                                                                                     |                                                                                  |                                                                                                                                                                                                                                      |                                                                                                       |
|  Lesson Narrative    | In this lesson, students will extend their learning from the prior lesson (Part 1) to apply it to real-world scenarios. Students will learn about corresponding parts in the real world and then use that knowledge to solve real-world problems that involve congruency. |                                                                                  |                                                                                                                                                                                                                                      |                                                                                                       |
|  Component Summary  | 5.10.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                  | Menu pricing Notice and Wonder activity ( <i>SE p. 277</i> )                     | 5.10.3 Collaboration (10-15 min.)                                                                                                                                                                                                    | Collaboration with solving real-world problems using corresponding parts ( <i>SE p. 278</i> )         |
|                                                                                                       | 5.10.2 Guided Instruction (10-15 min.)                                                                                                                                                                                                                                    | Identifying and using corresponding parts in the real world ( <i>SE p. 277</i> ) | 5.10.4 Your Turn (5-10 min.)                                                                                                                                                                                                         | Solving a real-world problem using corresponding parts ( <i>SE p. 279</i> )                           |
|                                                                                                       | 5.10.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                  | Self-reflection on the different concepts of this unit ( <i>SE p. 280</i> )      |                                                                                                                                                                                                                                      |                                                                                                       |
|  Resources         | Glossary                                                                                                                                                                                                                                                                  | Materials                                                                        | Additional Preparation                                                                                                                                                                                                               |                                                                                                       |
|                                                                                                       | <u>Key Terms</u> : N/A<br><u>Additional Terms</u> : N/A                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>(Optional) Markers</li></ul>               | <ul style="list-style-type: none"><li>Some of these problems can be more challenging to decipher than others. Work through each problem on your own prior to delivering this lesson to check for potential misconceptions.</li></ul> |                                                                                                       |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <br>Teacher Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|                                                                                                                | <ul style="list-style-type: none"><li>Students may struggle with reading or relating to the context.</li><li>Students may not understand which parts in congruent polygons are corresponding parts.</li><li>Students may not translate the question correctly using the provided information.</li></ul>                                                                                                                                                                                                                                                                |  | <ul style="list-style-type: none"><li>This lesson is the second of a two-part series. Each problem will require students to use word problem and reading comprehension skills in addition to the necessary mathematics. Consider a review of a word problem strategy prior to launching the lesson.</li><li>Students learned about interior angles sums in 8<sup>th</sup> grade Pre-Algebra. However, there may be students who do not remember. Consider a quick review (no more than 2 minutes) of decomposing polygons into triangles to find the sum of the angles.</li></ul> |  |
|                                                                                                                | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|                                                                                                                | <b>Notice and Wonder</b><br>5.10.1 Warm-Up asks students to determine things they notice and things they wonder about items on a menu, using the various values of different components.                                                                                                                                                                                                                                                                                                                                                                               |  | <b>MA.K12.MTR.7.1: Real-World Contexts</b><br>Throughout this lesson, students will be provided with complex problems about congruency through corresponding parts in real-world scenarios. Students will learn and practice decomposing these problems into manageable parts.                                                                                                                                                                                                                                                                                                    |  |
| <br>Differentiate Instruction | For a different entry point in the Warm-Up, provide specific scaffolding questions that will guide the students to what can be noticed when comparing the prices.<br>Prior to launching the 5.10.2 Guided Instruction, consider reviewing your chosen word problem strategy with students, modeling its use in the problems used during instruction.<br>The Wrap-Up provides students with a self-reflection on the concepts of the entire unit. Use their responses, coupled with their overall performance, to determine which students may need additional support. |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |

## 5.10.1 Warm-Up: Notice and Wonder

**Component Overview:** Students will complete a notice and wonder activity about menu prices.



### 5.10.1 Warm-Up: Notice and Wonder

A picture of a menu is shown.

| Burgers              | Chips   | Fries   |
|----------------------|---------|---------|
| Deluxe Burger        | \$6.90  | \$7.70  |
| Deluxe Cheese        | \$7.70  | \$8.45  |
| Bacon Burger         | \$7.95  | \$8.65  |
| Bacon Cheese         | \$8.25  | \$8.95  |
| Double Burger        | \$8.35  | \$9.30  |
| Double Cheese        | \$9.25  | \$9.95  |
| Double Bacon Cheese  | \$9.50  | \$10.25 |
| Cheese Garden Burger | \$8.25  | \$8.85  |
| Mushroom Swiss       | \$8.25  | \$8.95  |
| Jalapeno Pepper Jack | \$8.25  | \$8.95  |
| BBQ Western Bacon    | \$9.20  | \$9.85  |
| Pastrami Burger      | \$9.20  | \$9.85  |
| Power Pac            | \$10.45 | \$11.05 |
| Cheese Power Pac     | \$11.25 | \$11.70 |

- What do you notice about prices of the burgers with cheese?  
*Answers will vary. The burgers with cheese have higher prices, but not by the same amount.*
- What do you notice about ordering chips versus fries?  
*Answers will vary. Chips are less than fries, but also by different amounts.*
- There is a price difference between adding chips or fries with a double burger and adding chips or fries with cheese garden burger. Do you think this means that when you order a double burger you get more fries?  
*Answers will vary. I would not expect that this has anything to do with the amount of fries, however, maybe more people add fries to the double burger, so the company can make more money by charging more compared to the Cheese garden burger.*
- When you look at a menu, do you consider the price differences or just order what you would like to eat?  
*Answers will vary. I use to look at all prices, find the lowest few items, and select from them. Now I have started picking a few things I want to eat, then compare the cost.*



For a different entry point, provide questions to students that highlight specific things to notice. For example:

- What is the price difference between a Deluxe Burger and a Deluxe Cheese?
- What is the price difference between a Bacon Burger and a Bacon Cheese?
- What is the price difference between ordering a Deluxe Burger with Fries and ordering it with Chips?
- What is the price difference between ordering a Bacon Burger with Fries and ordering it with Chips?

## 5.10.2 Guided Instruction: Corresponding Parts in the Real World

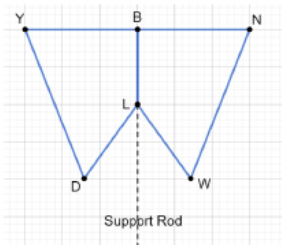
**Component Overview:** Students will receive guided instruction on how to identify and use corresponding parts in the real world.



### 5.10.2 Guided Instruction: Corresponding Parts in the Real World

Oscar and Aarna worked together to create a prototype obstacle for a drone obstacle course. They used grid paper to create their design as shown.

To be sure the design met regulations for the competition, they used the equations to input the design into a computer program. They discovered that some of their work disappeared, so they used the given equations knowing that  $YBLD \cong NBLW$ .



The perimeter of  $YBLD$  is 293 centimeters (cm).  $\angle YBL$  is a right angle.

$$\begin{array}{lll} YB = 75 \text{ cm} & WL = (3x + 10) \text{ cm} & m\angle LWN = (4k + 9)^\circ \\ m\angle DLB = (12k + 1)^\circ & WN = (6x + 5) \text{ cm} & LB = 50 \text{ cm} \\ & m\angle BYD = (5k + 8)^\circ & \end{array}$$

- What is the value of  $x$ ?  
 $75 + 6x + 5 + 3x + 10 + 50 = 293$   
 $x = 17$
- What is the length of  $WL$ ? Include units.  
 $WL = 3x + 10 = 61 \text{ cm}$
- What is the length of  $YD$ ? Include units.  
 $YD = WN = 6x + 5 = 107 \text{ cm}$
- What is the value of  $k$ ?  
 $5k + 8 + 90 + 12k + 1 + 4k + 9 = 360$   
 $k = 12$
- What is the measure of  $m\angle BYD$ , in degrees?  
 $m\angle BYD = 68^\circ$
- What is the measure of  $m\angle LDY$ , in degrees?  
 $m\angle LDY = 57^\circ$



When introducing the problem, ask students what they know about drones. Introducing this concept, help students understand and pay close attention to the key details that are presented within the context of the problem. Connect the instruction during this activity to what they did in the prior lesson so they can connect how these real-world problems can be solved using the same skills and knowledge.



When reviewing these questions during instruction, consider using markers or highlighters to connect where the information needed came from in the original image.

## 5.10.3 Collaboration: Corresponding Parts in the Real World

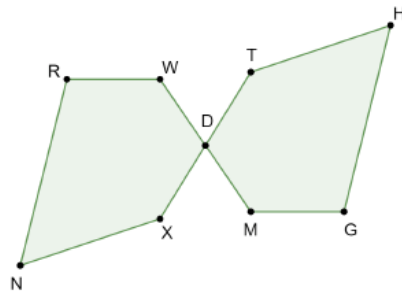
**Component Overview:** Students will work together to solve real-world problems using corresponding parts. For this activity, students should be allowed to work in pairs or small groups. Consider reviewing the context prior to releasing to students.



### 5.10.3 Collaboration: Corresponding Parts in the Real World

Work with your partner to complete the following.

French formal gardens use geometric shapes that repeat throughout the garden. Dajon is a landscape designer. He is designing a garden in the tradition of a French formal garden. While at the client's home, he wrote down some measurements, in feet (ft), and then sketched polygons  $RWDXN$  and  $DTHGM$  to represent the garden. In the tradition of the congruence of polygons found in French formal gardens, Dajon also wrote  $NRWDX \cong HGMDT$ .



The garden's measurements are given in the table.

| Garden Measurements  |                       |
|----------------------|-----------------------|
| $WD = (4x - 0.1)$ ft | $MD = 10.3$ ft        |
| $RW = 12$ ft         | $TH = (8x - 1.8)$ ft  |
| $DX = (3x + 3.4)$ ft | $HG = (10x - 1.3)$ ft |

- Determine the perimeter of the garden. Show your work.

$$WD = MD$$

$$4x - 0.1 = 10.3$$

$$x = 2.6$$

$$\begin{aligned}
 2(RW + WD + DX + NX + NR) &= \text{perimeter of the garden} \\
 2(12 + 10.3 + 3(2.6) + 3.4 + 8(2.6) - 1.8 + 10(2.6) - 1.3) &= \\
 \text{perimeter of the garden} \\
 2(12 + 10.3 + 7.8 + 3.4 + 20.8 - 1.8 + 26 - 1.3) &= \text{perimeter} \\
 \text{of the garden} \\
 2(77.2) &= \text{perimeter of the garden} \\
 \text{Perimeter of the garden is } 154.4 \text{ ft}
 \end{aligned}$$



Prior to releasing for this activity, it may be beneficial to read and review the written context. This may assist struggling readers with a head start in thinking through the context when working with their partner.



For question 1, students may use either polygon when writing the equation of the perimeter. When reviewing answers, allow both to be shared and validated as being correct answers.

### 5.10.3 Collaboration: Corresponding Parts in the Real World (continued)

Once the garden has been installed, Dajon will check the angles of the polygons.

|                                 |
|---------------------------------|
| $m\angle HTD = (5k - 6)^\circ$  |
| $m\angle XNR = 58^\circ$        |
| $m\angle NRW = (4k + 4)^\circ$  |
| $m\angle DMG = (6k - 20)^\circ$ |
| $m\angle MGH = 104^\circ$       |

2. Determine what measurement, in degrees, Dajon should use to check  $m\angle XDW$ . Show your work.

$$m\angle NRW = m\angle MGH$$

$$4k + 4 = 104$$

$$4k = 100$$

$$k = 25$$

$$m\angle NRW + m\angle RWD + m\angle XDW + m\angle DXN + m\angle XNR = 540^\circ$$

$$(4(25) + 4) + (6(25) - 20) + m\angle XDW + (5(25) - 6) + 58 = 540^\circ$$

$$104 + 130 + m\angle XDW + 119 + 58 = 540^\circ$$

$$m\angle XDW + 411 = 540^\circ$$

$$m\angle XDW = 129^\circ$$



For question 2, students learned about interior angles sums in 8<sup>th</sup> grade Pre-Algebra. However, there may be students who took Algebra 1 instead of Pre-Algebra, potentially missing that content. Consider a quick review (no more than 2 minutes) of decomposing polygons into triangles to find the sum of the angles.



Facilitation questions to consider:

- How do you know which measurement Dajon should use to check that angle?
- Why aren't other angles good options for checking that angle?

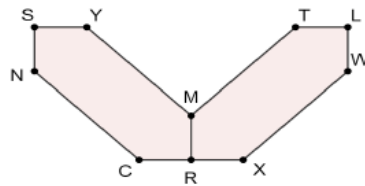
## 5.10.4 Your Turn!

**Component Overview:** Students will practice solving a real-world problem using corresponding parts. For this activity, allow students to work independently to determine their level of understanding of the instruction provided in the prior activities. After a given set of time, review the answers as a whole group.



### 5.10.4 Your Turn!

1. Mireia is learning how to make stained glass in art class. She has created a design and labeled the vertices to help her keep track of the measurements. Her design is shown.



Mireia's design was created so that  $RMYSNC \cong RMTLWX$ ,  $\overline{CN} \cong \overline{MY}$ , and  $\overline{CR} \cong \overline{RM} \cong \overline{YS} \cong \overline{SN}$ . Mireia needs to determine how much solder she needs. Solder is used along the perimeter, in inches (in.), of the design to set the glass in place.

Use the given measurements, in inches, to determine the amount of solder.

$$WX = 9x - 2.3$$

$$YM = 6x + 1.3$$

$$MR = 3x - 0.6$$

$$WX = YM$$

$$9x - 2.3 = 6x + 1.3$$

$$3x = 3.6$$

$$x = 1.2$$

$$YM = TM = WX = NC = 6(1.2) + 1.3 = 8.5$$

$$MR = SY = SN = CR = XR = LW = TL = 3(1.2) - 0.6 = 3$$

$$\text{Perimeter} = YM + YS + SN + NC + CR + RX + XW + WL + LT + TM + MR$$

$$\text{Perimeter} = 8.5 + 3 + 3 + 8.5 + 3 + 3 + 8.5 + 3 + 3 + 8.5 + 3$$

$$\text{Perimeter} = 55 \text{ inches of solder}$$



The time allotted for this activity would be a great opportunity to pull small groups of students who may need additional support based on their performance in the prior activity and prior lesson.



Students may wonder how to handle the line segment MR. Consider monitoring students to ensure that they understand it as a shared side between the two pieces of stained glass and that solder would not need to be placed there twice.



## 5.10.5 Wrap-Up: Reflection

**Component Overview:** Students will complete a self-reflection on their understanding of the different concepts of this unit. This activity should be done independently to get student-level individual reflection data.



### 5.10.5 Wrap-Up: Reflection

1. Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentence by checking all that apply.

*Answers will vary.*

I don't understand

- ☐ SSS
- ☐ SAS
- ☐ ASA
- ☐ AAS
- ☐ HL
- ☐ CPCTC

... YET.

- ☐ Naming congruent parts
- ☐ Paragraph proofs
- ☐ Two-column proofs
- ☐ Flowchart proofs
- ☐ Finding values in congruent polygons

do(does) NOT make sense ...  
YET.

I can explain

- ☐ SSS.
- ☐ SAS.
- ☐ ASA.
- ☐ AAS.
- ☐ HL.
- ☐ CPCTC.
- ☐ naming congruent parts.
- ☐ finding values in congruent polygons.



*This activity provides students with a self-reflection on the concepts of the entire unit. Use their responses, coupled with their overall performance, to determine which students may need additional support before the end of unit assessment.*